Rohit Kota

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Abu Dhabi, UAE | UAE Golden Visa (Valid till 2035) | Open to relocation & travel

SUMMARY

Computer Vision and AI Platform Engineer who turns HSE and operational requirements into AI vision models and production alerts that hold up in the field. On RASED, ADNOC's computer-vision safety platform, deployed to 70+ sites across eight ADNOC business units in just over a year, fine-tuning PyTorch detectors on site footage, deploying quantized TensorRT models to GPU edge nodes, and prototyping multimodal LLM agents to reduce false alarms. Owns use cases end to end, from requirements and model validation to production safety events, and has demonstrated the platform live at GITEX and ADIPEC.

Earlier, at Rakuten Symphony, built the storage layer supporting AI workloads, including garbage collection and S3 APIs for an erasure-coded object store used for training datasets and model checkpoints. The platform replaced MinIO and contributed to more than $4M in annual licensing savings, alongside core features for a Kubernetes automation platform. Four years of hands-on engineering experience across PyTorch, TensorRT, Kubernetes, C, Python, and PostgreSQL, spanning production AI, distributed systems, and cloud-native infrastructure.

EDUCATION

College of Engineering, Pune — Pune, India
Bachelor of Technology in Computer Engineering — *Aug 2018 – Jun 2022*

SKILLS

- **Languages**: Python (asynchronous services with Sanic and asyncpg, REST APIs, backend development), C, JavaScript, SQL, Bash; working knowledge of C++ and Go
- **Computer Vision, ML & Generative AI**: PyTorch, TensorFlow/Keras, OpenCV, scikit-learn, NLTK, CNN and transformer (ViT) detectors, object detection and tracking, dataset annotation and augmentation, evaluation metrics (mAP, precision/recall), hyperparameter tuning, model optimization (quantization, TensorRT), multimodal LLMs and vision-language models (VLMs), prompt engineering and fine-tuning, responsible AI practice (false-alarm and bias analysis, human review before escalation), MLOps (MLflow experiment tracking, model versioning and release)
- **Platforms & DevOps**: Kubernetes (bare-metal and cloud), Docker, Helm, CI/CD (Jenkins, Azure DevOps), GitOps, Longhorn, Veeam Kasten, NGINX, Linux (Ubuntu), PostgreSQL/Patroni, Redis, MQTT, Sanic, Flask, DAG workflow orchestration (Argo Workflows and Kubeflow Pipelines), logging and monitoring
- **Cloud & Enterprise Integration**: Azure (Entra ID, API Management, DevOps, Container Registry), AWS (EKS, S3, IAM), Keycloak, LDAP/SSO, Power BI, SAP HSE, REST APIs and webhooks, identity and access management, cloud security (OAuth2/JWT, TLS, encryption), secure API integration
- **Tools**: Git, GDB, ASan, K9s, Postman, Wireshark, Jenkins, Selenium

WORK EXPERIENCE

Trutech (Fogsphere delivery partner) — Abu Dhabi, UAE
Senior AI Engineer, RASED – ADNOC Video Analytics Platform (Fogsphere) — *Aug 2025 – Present*

Core engineer on RASED, ADNOC's real-time AI safety platform that runs computer-vision models on live CCTV to detect unsafe acts and conditions, deployed across eight ADNOC business units: Drilling, Onshore, Offshore, Gas, Sour Gas, Distribution, Borouge, and ADNOC HQ. Work across the platform end to end, from requirements with ADNOC HSE teams and site solution design through computer-vision use-case engineering and validation, backend integrations, Kubernetes and edge deployment, and production troubleshooting, serving as the day-to-day technical interface between ADNOC, Fogsphere engineering, and e& (Etisalat) field teams.

Computer Vision & Solution Engineering

- Own **requirements for new safety use cases** with ADNOC site managers and HQ HSE teams, turning operational risks into detection definitions, acceptance criteria, and alert routing so every camera has a measurable purpose before deployment.
- Design **site-level solutions**: camera coverage per use case, edge sizing by camera count and GPU load, bandwidth plans for event sync and training clips, network and firewall rules, and HA main/backup edge pairs.
- Build production **computer-vision use cases** combining object detection, person tracking, and zone and dwell-time rules to catch PPE violations, red-zone and line-of-fire breaches, man-down, fire and smoke, and vehicle–personnel proximity.
- Fine-tune and evaluate **detection models in PyTorch** on ADNOC site data, building annotation, augmentation, and preprocessing pipelines and tracking experiments in MLflow, raising mAP and recall under site conditions such as dust, glare, and night footage.

- Optimize **models for edge GPU inference** by calibrating confidence thresholds and regions of interest per camera, exporting quantized TensorRT engines, and releasing versions to edge nodes through the model store, cutting false alarms while maintaining recall.
- Prototype **multimodal LLM (vision-language model) agents** that generate plain-language descriptions of detected events and suppress false alarms before escalation, through prompt design and fine-tuning on annotated site clips, benchmarking against the detector-only baseline to raise alert precision.

Platform, Backend & DevOps

- Engineer the **event pipeline and enterprise integrations**, including ADNOC event schemas, PostgreSQL reporting views for Power BI, and SAP HSE incident submission through Azure API Management, so safety events land in ADNOC systems without manual re-entry.
- Integrate **identity and alerting**: Microsoft Entra ID single sign-on with group-to-role mapping via Keycloak, email through the corporate relay, and SMS through Azure API Management, resolving provisioning and access issues for 800+ users.
- Deploy and operate the **central platform** on an 8-node NVIDIA L40 GPU Kubernetes cluster using Helm and GitOps upgrades with rollback, supporting a 99% availability target across 2,400 cameras and 200 sites.
- Partner with **DevOps and ML teams on platform architecture**, microservice scaling, performance, capacity planning, and release design so upgrades and model releases do not interrupt live detection.
- Commission and run **GPU edge inference nodes** at 70+ drilling rigs, offshore facilities, gas plants, and fuel distribution sites, from staging to camera stream integration and main/backup HA pairs, keeping detection running during VSAT and SD-WAN outages with store-and-forward.
- Run the **fleet-wide edge software release cycle**, planning Docker-based bundle rollouts with Fogsphere engineering and scheduling installs with e& field teams in batched windows, keeping remote desert and offshore rigs on current software versions with fewer site visits and no customer disruption.

Production, Validation & Customer Delivery

- Troubleshoot **production incidents on live rigs** independently, tracing missed or false alerts across camera streams, edge GPU load and connectivity, workflow configuration, and backend services to the root cause, and driving permanent fixes with Fogsphere engineering.
- Resolve **edge hardware and GPU performance issues** with e& field engineers, from GPU throttling and server health to cameras below minimum spec, and tune GPU load and CUDA settings with Fogsphere to keep each node at full camera capacity.
- Validate **releases and fixes against operational requirements**, running UAT with site HSE teams, replaying positive and negative scenarios on live cameras, confirming event generation and alert routing, and signing off versions for production.
- Serve as **ADNOC's day-to-day technical counterpart** across HSE, IT, engineering, and field teams, running requirement and review sessions, converting defects and change requests into delivery plans, training site teams and administrators, and managing expectations on live-rig priorities.
- Demonstrated the **Fogsphere platform at GITEX and ADIPEC**, running live computer-vision demos and walking customers, partners, and industry visitors through detection use cases, event handling, and architecture, adjusting depth for technical and executive audiences.

Rakuten Symphony (Robin.io) Pune, India
Associate Software Engineer | C, Python, Kubernetes, PostgreSQL *Jun 2022 – Sep 2024*

Object Storage

- Built core subsystems of an in-house **S3-compatible object store for AI datasets, model checkpoints, and artifacts** that replaced MinIO across production Kubernetes clusters, cutting third-party licensing costs by over $4M a year.
- Packaged the entire store into a **single-command Kubernetes deployment** using Helm charts with automatic rollback and disk-discovery DaemonSets, bringing up three 3-node synchronously replicated Patroni PostgreSQL clusters.
- Designed the **distributed garbage-collection system**, combining a transactional PostgreSQL write-ahead delete queue with at-least-once dispatch and a multithreaded disk scanner that checks chunks against metadata, reclaiming orphaned space even through crashes and restarts.
- Implemented **four S3 object APIs** to AWS spec on liberasurecode Reed-Solomon (4+2) striping: zero-copy CopyObject using link(2) hardlinks, HeadObject, ListMultipartUploads, and RFC-7233 ranged GETs over block-aligned O_DIRECT reads, the byte-range pattern ML data loaders use on sharded datasets.
- Owned the **license management system**, issuing RSA-4096-encrypted keys through a Flask service with Evaluation, Production, and Community tiers, per-cluster capacity limits up to 1 PiB, and expiry checks.
- Developed **cross-site configuration sync** for multi-cluster replication, a REST API that registers remote peer clusters and keeps their configuration consistent in PostgreSQL, surviving database crashes and peer outages through durable work queues and bounded retries.
- Wrote and ran **failure-mode and HA test plans** covering pod, node, and filesystem-full failures, network partitions, and Patroni PostgreSQL failover, verifying end-to-end S3 availability on bare metal with the AWS CLI.

MDCAP - Multi Data Center Automation Platform (metal-to-service orchestration engine behind Automation Studio)

- Built core features of a **cloud-native automation platform for telco infrastructure fleets** (async Python/Sanic, PostgreSQL, Kubernetes), running declarative DAG workflows for Day-0 to Day-2 provisioning, upgrades, and migrations across bare-metal servers and Kubernetes clusters, executed on containerized worker pods.
- Shrank the platform's **core container image by 90%** (1.78 GB to 180 MB) with a multi-stage build shipping PyInstaller single-file binaries on a minimal Alpine base instead of a full Python runtime and source tree, cutting the attack surface to close a customer security escalation.
- Extended the **declarative workflow DSL and DAG orchestration engine**, the same class of engine as Argo Workflows and Kubeflow Pipelines, with serial and parallel loop unrolling and task grouping, then used it to build a 55-step resumable workflow migrating telco infrastructure from Cisco VIM to Robin's cloud-native platform.
- Cut **fleet dashboard load time at scale** by collapsing per-resource PostgreSQL count queries into one PL/pgSQL aggregation that returns planner estimates instead of exact counts above 10,000 rows, backed by a 5-minute incremental stats snapshot.
- Implemented **least-privilege RBAC** by encoding per-resource-type CRUD bitmaps in JWT claims and enforcing them in API middleware by HTTP verb, creating tiered Admin, Moderator, and Viewer roles across 20+ resource types for 1,000+ clients.
- Developed **secure browser-based access to servers and BMCs** by customizing the Wetty web terminal behind an Nginx WebSocket proxy, with short-lived per-session tokens, no credentials in process arguments, and on-demand ipmitool power control for out-of-band management.
- Added **hybrid-cloud fleet support by onboarding Amazon EKS clusters** as managed resources alongside on-prem Kubernetes, extending the shared resource model with an EKS schema, IAM-based authentication, and lifecycle workflows for cluster operations and network services.

INTERNSHIP

Algoanalytics Pune, India
Research Intern – Cloud *Jul 2021 – Aug 2021*

- Evaluated **cloud AI/ML platforms for hosting AlgoAnalytics' models**, comparing managed training and inference services, GPU compute options, storage, and pricing models across AWS, Azure, and Google Cloud, and recommended deployment targets by cost and workload fit.
- Mapped the **architectures and technology stacks of competing AI products** in finance, healthcare, and retail, from data pipelines and model serving to cloud dependencies and APIs, and turned the findings into a gap analysis and deployment recommendations for the product team.

ACADEMIC PROJECTS

- **Xv6 Scheduling Algorithms**: extended the MIT xv6 operating system with new system calls and scheduling algorithms (FCFS, Priority Based Scheduling, Lottery Scheduling, MLFQ), comparing and analyzing their performance.
- **Linux Shell**: developed a command-line interpreter in C demonstrating process control and inter-process communication on UNIX-like systems, with pipes, I/O redirection, shell built-ins, and job control.
- **HTTP Server**: implemented a multithreaded HTTP/1.1-compliant web server in Python supporting GET, POST, PUT, HEAD, and DELETE, cookies, headers, non-persistent connections, concurrent clients, leveled logging, file permissions, server configuration, connection limits, and stop/restart.
- **Fake News Detection**: built a Python classifier over a labeled corpus using NLTK preprocessing (stopword removal, stemming), bag-of-words and TF-IDF features, and Naive Bayes, Logistic Regression, SVM, SGD, and Random Forest models, selecting the best by F1 score and accuracy.
- **Hospital Management System**: developed a MySQL-based system to manage patient records and hospital workflows.
- **Grep**: implemented a multithreaded grep in C that searches multi-gigabyte files faster than the original single-threaded grep.
- **Career Grip**: created a MERN stack web app for managing job applications and recruiting workflows.

PUBLICATIONS

Electronic Health Record Management using Hyperledger Fabric IEEE ICCUBEA 2022
ieeexplore.ieee.org/document/10010988
6th International Conference on Computing, Communication, Control and Automation (ICCUBEA) *First author, Aug 2022*

- Designed a **patient-controlled electronic health record system on Hyperledger Fabric**, a permissioned ledger where patients grant and revoke doctors' access and every update is recorded immutably, so records follow the patient across hospitals.
- Showed how the design meets the **security, integrity, confidentiality, and privacy requirements** of sharing sensitive medical data, using certificate-based identities, membership services, and fault-tolerant consensus in place of a trusted central party.